Constraints in design and operation of autonomous driving

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Acronyms

- Autonomous driving vehicles (ADVs)
- Computing engine (CE): refers to the main processing engine of autonomous driving system
- LIDAR: Light Detection and Ranging
- Deep neural network (DNN)

Interest in Autonomous Vehicles is Growing

MIT

Google

MobileEye (Intel)

NVIDIA

Two broad approaches for ADVs

1. **Cloud-based:** Cars share sensor data with cloud. Cloud performs complex processing and advises cars on how to navigate the environment
   - Cloud has more storage & processing power
   - Has aggregate view of city/traffic, can report accidents, congestion, etc.
   - Transmission failure/delay, security issues
   - CLOUD CAN NEVER CONTROL CAR STEERING!
2. **Local processing using accelerators**
   - Highly reliable and secure
   - Required for real-time action!

In this PPT, we focus on local-processing based ADVs

6 Levels of Automation

0	No automation	Human driver must complete all driving tasks even with warnings from vehicles
1	Driver assistance	Auto-drive system shares steering and acceleration responsibility with human driver under limited driving conditions (e.g., high speed cruising). Diver handles the remaining driving tasks (e.g., lane change)
2	Partial automation	Auto-drive fully controls the steering and acceleration of vehicles under limited driving conditions, and human driver performs remaining driving tasks
3	Conditional automation (**eyes off**)	Auto-drive system handles all driving tasks under limited driving conditions, and expects that the human driver will respond to requests to intervene (i.e., resume driving)
4	High automation (**mind off**)	Auto-drive system handles all driving tasks under limited driving conditions even if the human driver does not respond to requests to intervene
5	Full automation (**driver off**)	Auto-drive system takes full control of all driving tasks under all driving conditions that can be managed by a human driver

Constraints in design and operation of ADVs

- Performance
- Predictability
- Storage
- Thermal
- Power
- Others e.g. mechanical (ignored in this PPT)

We now discuss first 5 constraints.

Performance constraints (1 of 2)

- Reaction time of ADV is determined by two factors

1. **Processing latency:** how fast the system can react

2. Frame rate: determines how fast the real-time sensor data can be fed into the processing engine

- Fastest action by a human driver takes 100-150 ms
  - E.g., A typical driver takes 0.96 s to release accelerator, 2.2 s to reach maximum breaking, and 1.64 s to begin steering to avoid an accident
- ADV should react faster than that
- => ADV processing latency <= 100 ms

Performance constraints (2 of 2)

- Further, ADVs need to frequently update their "understanding" to keep up with the continuously changing real-time traffic condition
- => Frame rate should be high if traffic condition changes quickly

- **Summary:** ADV should process traffic conditions with 100ms latency at a frequency of at least once every 100 ms

Predictability constraint

- Different phases of computation in ADVs show large performance variability depending on computation load
- **Predictability constraint:** tail latency (e.g., 99th-, 99.99th- percentile latency) should be used as the metric to evaluate the performance
- This captures the stringent predictability requirement

Storage constraint

- GPS: does not provide necessary precision (0.1m precision required for keeping the vehicle in a lane)
- => Use "prior-map" based localization for high precision
  - here, the surrounding view is transformed into feature descriptions to map the feature points stored in the prior map to identify vehicle location
- To avoid need of accessing cloud for prior-map, store it on vehicle itself.
- Challenge: requires huge storage, e.g., 41TB for map of whole US
- **Storage constraint:** tens of TBs

Thermal constraint (1 of 2)

- It has two aspects:
- (1) temperature of space where CE is kept, should be within system's operating range
- (2) heat generated by computing system should have low impact on vehicle (e.g., not heat up the engine)
- Outside the passenger cabin, temperature can get quite high
- => CE should be kept in climate-controlled passenger cabin

Thermal constraint (2 of 2)

- But then, passenger has to bear extra heat and temperature due to CE
- A CE consuming 1kW power can raise temperature by 10°C in a minute without additional cooling in passenger cabin
- => extra AC is required with CE in cabin

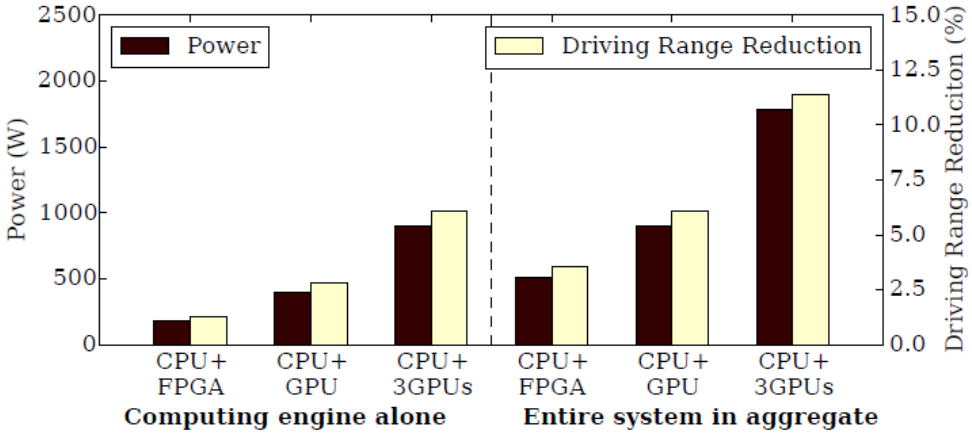
- **Summary:** keep CE in passenger cabin and add extra cooling capacity for removing heat and regulating temperature

Power constraint (1 of 2)

- Additional power consumption of ADV reduces driving range of vehicle and also fuel efficiency (miles per gallon)
- Total power consumption = CE+Storage+Cooling
- Power of CE depends on its type, e.g., CPU/FPGA/GPU

Reduction in driving range of a Chevy Bolt with increased power needs

Entire system = CE+Storage+cooling



Power constraint (2 of 2)

- Findings: CE contributes only ~half of power and range-reduction. Storage and cooling contribute remaining half.

- **Summary:**
- Total power consumption = CE+Storage+Cooling
- A power-hungry ADV can greatly degrade fuel efficiency.

References

- “The Architectural Implications of Autonomous Driving: Constraints and Acceleration”, ASPLOS’18